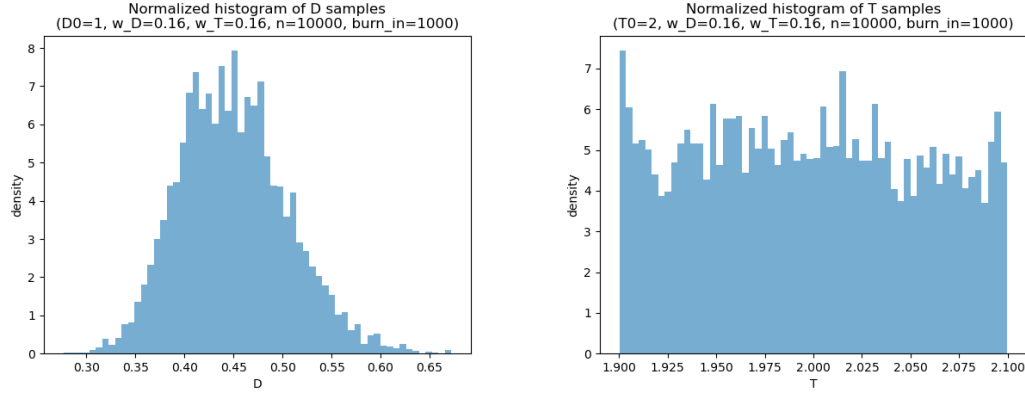


## Compute 3

### Exercise 1



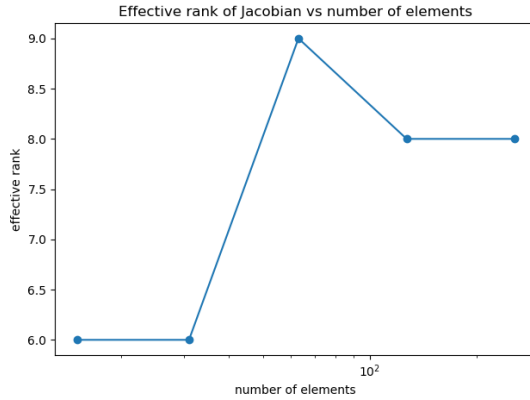
### Exercise 2

Given a possibly non-constant  $D : [0, L] \rightarrow \mathbb{R}^+$ , we compute the noise-free measurements  $(u_D(x_i, T))_i$  via the finite element method.

In particular, we choose some number of elements  $n$ , cut the interval  $[0, L]$  into the elements  $(x_i, x_{i+1})$  for  $i = 1 \dots n$ , and discretize  $D()$  to be a vector of length  $n$ ,  $(D_i)_{i=1}^n$ , where  $D_i$  is  $D()$  evaluated at the midpoint of the  $n$ -th element. We then perform the finite element method as usual.

The Jacobian of the forward map  $F : D() \mapsto (u_D(x_i, T))_i$ , is computed via a finite difference approximation.

Below we plot the number of elements used in the FEM vs the effective rank of the Jacobian.



As the number of elements increases, the effective rank of the Jacobian increases initially but then plateaus at around 8 for more than 127 elements.